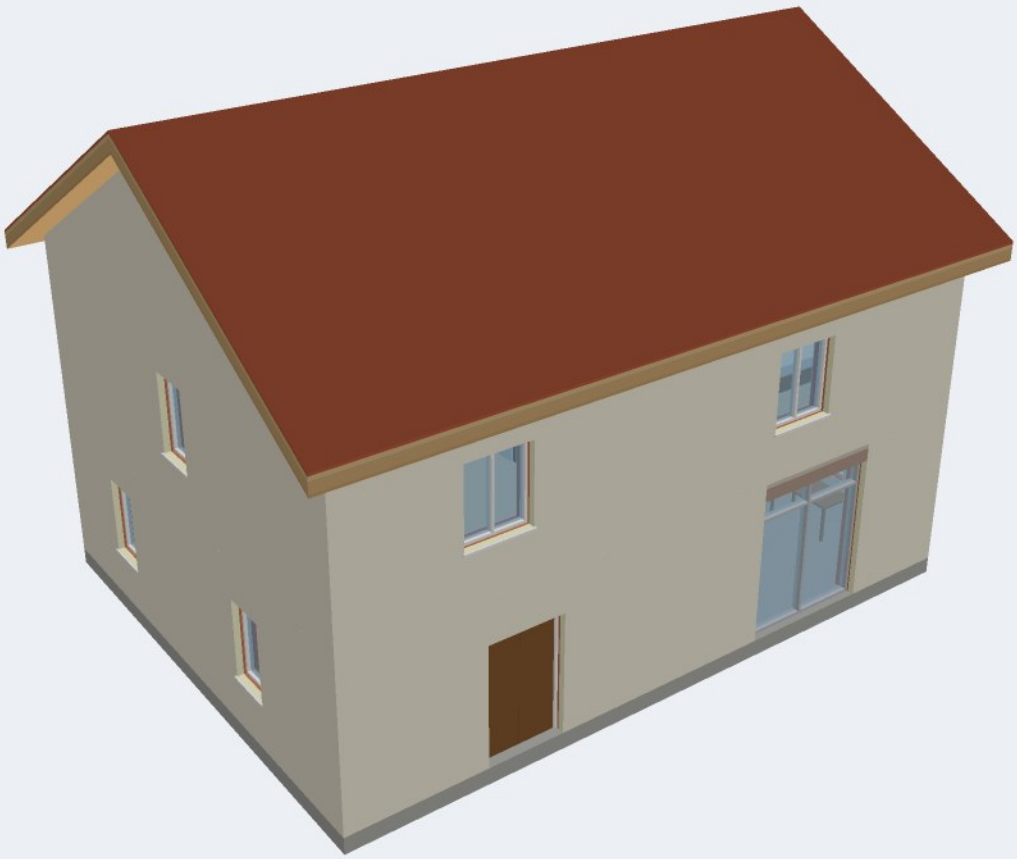
BIMopen user manual



Antonio Sanna
BIMopen
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BIMopen user manual

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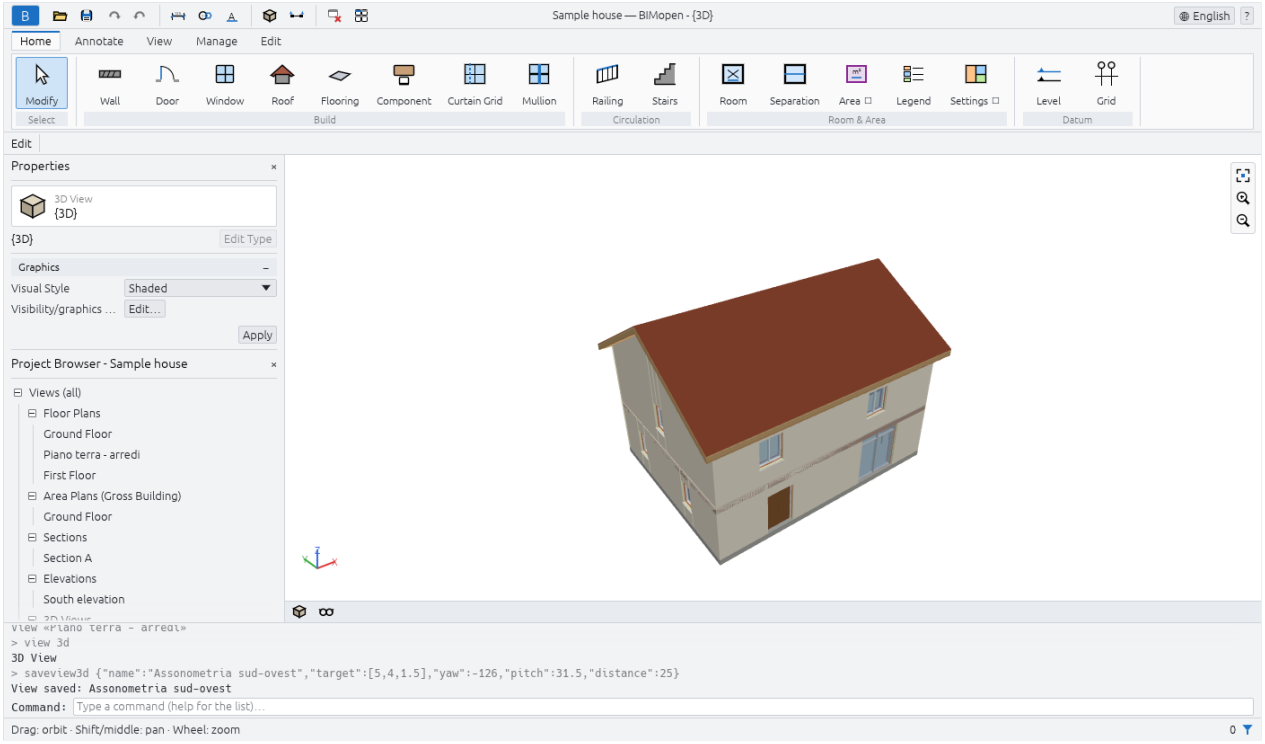
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BIMopen user manual

BIMopen is an open-source application for building information modelling (BIM). You model the building with walls, doors, windows, floors, roofs, stairs and rooms, and the drawings come from the model: plans, sections, elevations, 3D views, schedules and sheets are views of the same model and change together when you edit it.

BIMopen runs on the desktop (Linux, Windows, macOS) and in a web browser, and it can be driven from a command line and by AI agents over MCP. Its interface speaks six languages: Italian, English, German, French, Spanish and Arabic, the last one written from right to left.



How this manual is organised

Chapter	What it covers
Getting started	Installing and starting BIMopen, the sample project, and a first small house from an empty project to a sheet
The interface	The application menu, the quick access toolbar, the ribbon, the options bar, Properties, the Project Browser, the drawing area, the status bar, the command line and how to move around in the views
Projects and files	Opening, saving, exporting and importing projects
Selecting and modifying	Selecting elements and the Modify tools
Walls, doors and windows	The building envelope and its openings
Floors, roofs, stairs and components	Floors, roofs, stairs, railings, curtain walls and placed components
Views and graphics	Plans, sections, elevations, 3D views and how they are drawn
Annotation and 2D details	Dimensions, text, tags, rooms, areas and 2D details
Sheets, schedules and printing	Sheets, schedules, printing and project settings
Command line and MCP	Typing commands, `bimopen-cli` and the MCP server for agents
Keyboard shortcuts	Every key and typed shortcut
Troubleshooting	What to do when something does not work as expected

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The chapters from *Projects and files* to *Sheets, schedules and printing* are being written area by area: each one is completed when the features it describes have been checked one by one through the interface. Until then they list what they will contain.

Conventions

- Names of the interface are written in **bold**, as they appear on the screen in English: **Home** tab, **Wall** button, **Properties** palette.
- A path through the interface is written with arrows: **Home ▸ Build ▸ Wall** means the **Home** tab of the ribbon, its **Build** panel, its **Wall** button.
- Keys are written like `Ctrl+Z`; a shortcut typed at the keyboard, letter by letter, is written in capitals, like `WA`.
- Commands typed at the command line, and their output, are written `like this`. Points are `x,y` in metres.
- Notes, tips and warnings are set apart:

> **Note**
>
> A note adds something worth knowing.

> **Tip**
>
> A tip shows a quicker or better way.

> **Warning**
>
> A warning tells you about something that can lose work or give a wrong result.

What this manual describes

This manual describes BIMopen as it is today, and only what has been tried in the program. When a feature is not described yet, it may still exist: the roadmap *(https://atsanna.github.io/bimopen-site/roadmap.html)* lists what is available and what is planned. The whole manual is also a PDF, one per language: the last entry of the navigation, **PDF**, downloads it.

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Getting started

This chapter shows how to start BIMopen on the desktop and in the browser, what the sample project contains, and how to model a first small house: four walls, a door, a window, a floor, a roof, a 3D view and a sheet with the plan on it. It takes about ten minutes.

Running BIMopen on the desktop

BIMopen is built from its source code with Rust, version 1.90 or later. On Linux the graphical interface also needs the development files of xkbcommon, Wayland and GTK 3.

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev   # Debian, Ubuntu
cargo run --release -p bimopen -- --sample                             # opens the sample house
```

The program is `bimopen`. It accepts a file to open and one option:

Command	What it opens
`bimopen`	A new, empty project
`bimopen --sample`	The sample house
`bimopen house.bimopen`	The project saved in `house.bimopen`

You can also open a project by dragging a `.bimopen` file from the file manager into the window.

Language and theme

The first time, BIMopen speaks the language of the operating system when it is one of its six languages (Italian, English, German, French, Spanish, Arabic), and English otherwise. To change it, use the language menu at the right end of the title bar (the globe with the language's name), or **Application menu ▸ Options ▸ Language**. The interface changes at once.

In **Options** you also choose the **Theme**, **Light** or **Dark**, and whether the background of the views is inverted (**Invert background color**: dark views with light lines). Choosing the dark theme also inverts the background; you can turn it back off. BIMopen remembers the language and the theme for the next time.

Running BIMopen in the browser

The web version is the same program, compiled for the browser. Open atsanna.github.io/bimopen-site/app/ *(https://atsanna.github.io/bimopen-site/app/)*: it starts with the sample house. Nothing needs to be installed, and the project stays in your browser: it is not sent anywhere.

The address accepts these options, which can be combined with `&`:

Address	Effect
`app/?lang=de`	Opens in German for this visit (`it`, `en`, `de`, `fr`, `es`, `ar`)
`app/?empty`	Starts with a new, empty project instead of the sample
`app/?webgl`	Draws with WebGL2 even where WebGPU is available

The browser must support WebGPU or WebGL2. In the browser some things work differently:

- **Open…** cannot show a file dialog: drag a `.bimopen` file into the page to open it;
- **Save** and the exports download the file, with the name of the project.

To build the web version yourself:

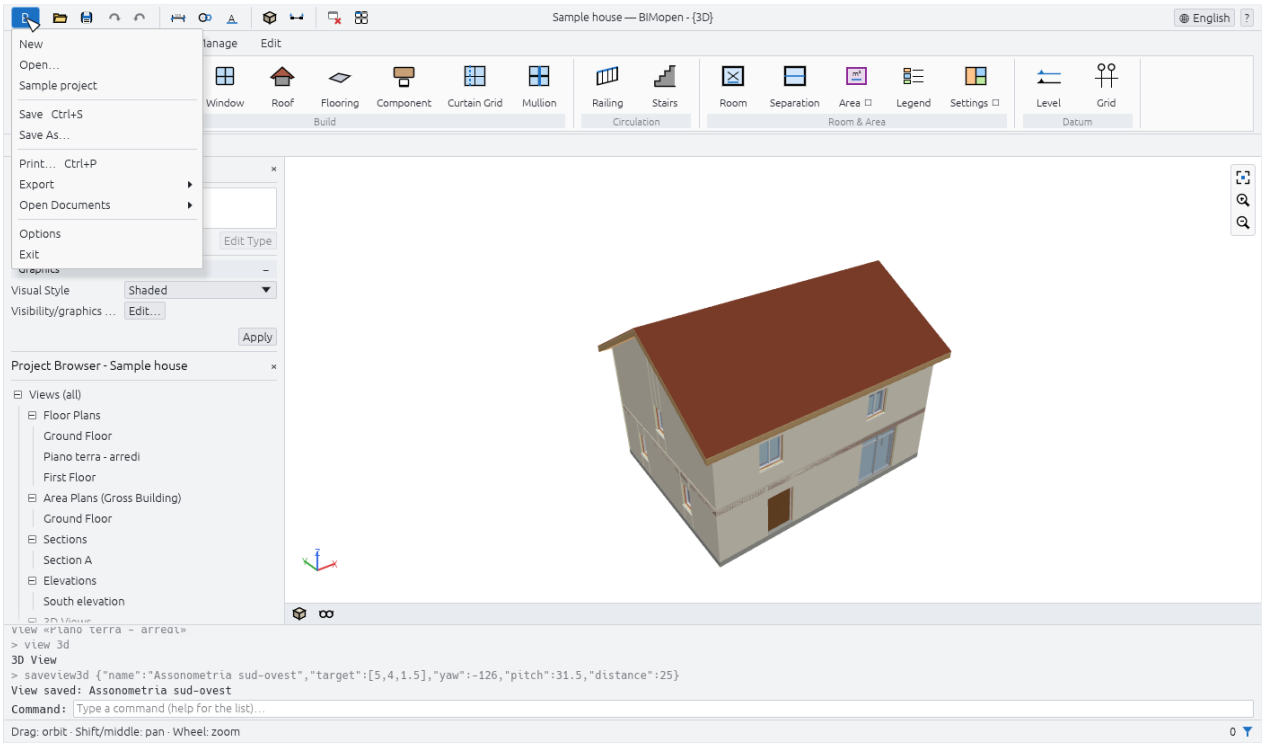
```
cargo xtask web                          # builds dist/web
python3 -m http.server 8080 -d dist/web  # then open http://localhost:8080
```

`cargo xtask web` needs the `wasm32-unknown-unknown` target of Rust and the `wasm-bindgen` command of the version written in `Cargo.lock`; with binaryen's `wasm-opt` installed, the result is smaller.

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The sample project

The sample house shows most of what BIMopen can do. Open it with **Application menu ▸ Sample project**, with `bimopen --sample`, or by typing `sample` at the command line.



It contains:

- two levels, each with its floor plan, and five grid lines;
- fifteen walls of four types, with layers, joined at the corners;
- six doors and twelve windows, tagged in the plans;
- two floors, a pitched roof, a stair with its railings and a storefront curtain wall;
- furniture and plumbing fixtures, eight rooms coloured by use, with a legend;
- a section, an elevation, an area plan, a saved 3D view and a drafting view with a wall detail;
- dimensions, text notes and the schedules of walls, doors, windows, rooms and materials.

> **Tip**
>
> The sample is a good place to try a tool before using it on your own project: a new sample is always one click away, and nothing you change in it is saved unless you save it.

Your first house

The steps below start from an empty project and end with a sheet. Every step names the button to click; the interface chapter explains each part of the window.

1. Start a new project

1. Click the blue **B** button at the left end of the title bar: the **application menu** opens.
2. Click **New**.

The new project has two levels, **Ground Floor** and **First Floor**, a floor plan for each, and the default 3D view. It has no walls yet. Its wall, door, window, floor and roof types are ready to use.

2. Open the ground floor plan

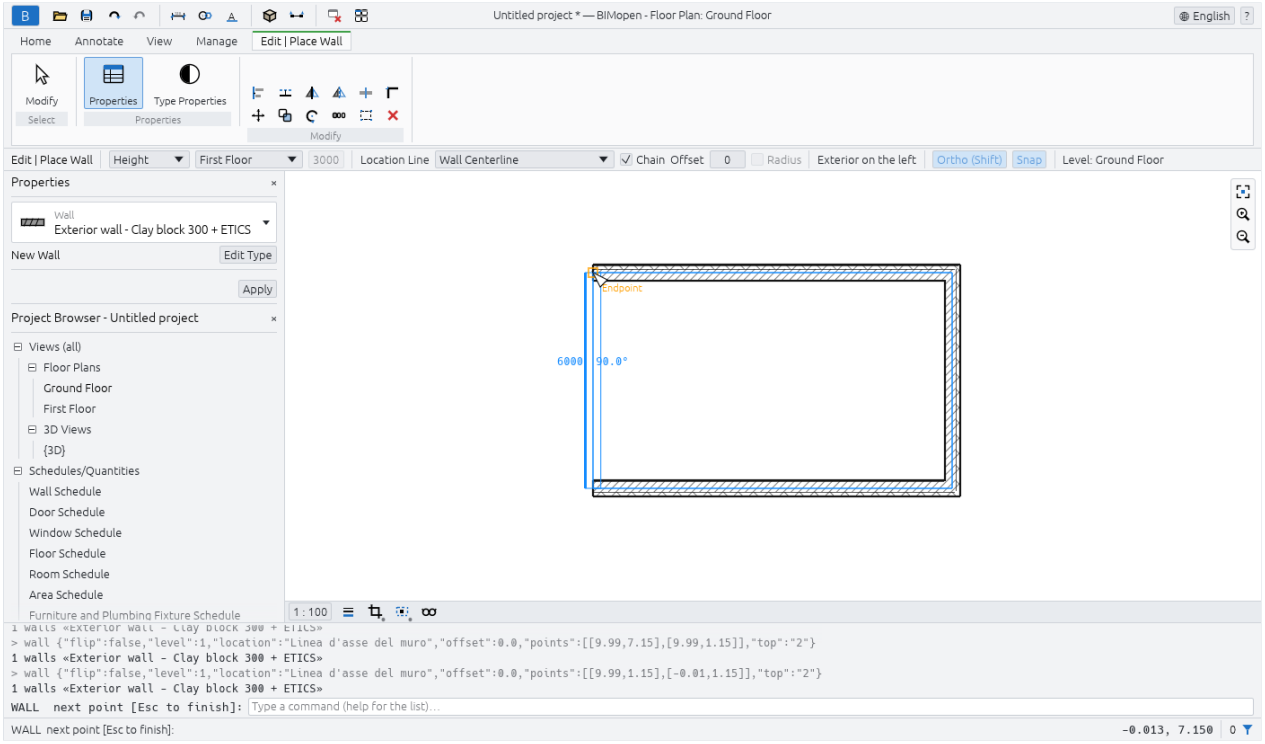
In the **Project Browser**, under **Views (all) ▸ Floor Plans**, double-click **Ground Floor**. The plan opens in the drawing area and the title bar shows **Floor Plan: Ground Floor**.

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> **Note**
>
> A single click on a view in the Project Browser shows its properties in the **Properties** palette without opening it. A double-click opens it.

3. Draw four walls

1. Click **Home ▸ Build ▸ Wall**. The ribbon switches to the contextual tab **Edit | Place Wall**, the options bar shows the settings of the wall, and **Properties** shows the type of the next wall, **Exterior wall - Clay block 300 + ETICS**.
2. On the options bar, turn on **Ortho (Shift)**: the walls stay horizontal or vertical.
3. Click in the plan where the first corner goes. The command line asks for the next point: `WALL next point [Esc to finish]:`.
4. Click the second, third and fourth corners. While you move the mouse, the length and the angle of the wall being drawn are shown next to it.
5. Click the first corner again: the four walls are joined at the corners.
6. Press `Esc` to stop the chain, and `Esc` again to end the tool.



> **Tip**
>
> You can draw the same walls by typing their corners in metres: `wall 0,0 10,0 10,6 0,6 c`. The final `c` closes the chain. See Command line and MCP.

4. Add a door and a window

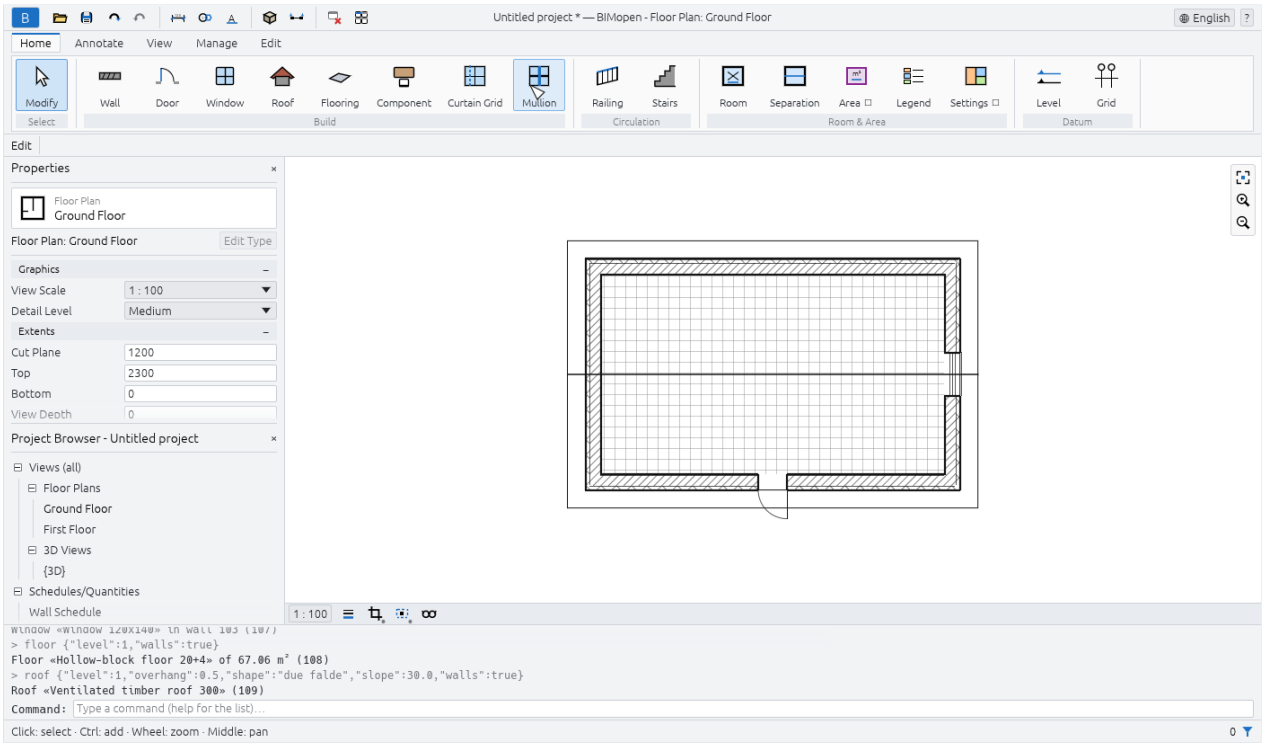
1. Click **Home ▸ Build ▸ Door**, then click on a wall where the door goes. The door is placed in the wall, which is cut around it. The command line confirms it: `Door «Single swing door 80x210» in wall …`.
2. Press `Esc`: the tool ends and the ribbon returns to the **Home** tab.
3. Click **Home ▸ Build ▸ Window**, click on another wall, and press `Esc`.

A door or a window can only be placed in a wall: after a click away from the walls, the command line answers `no wall at that point` and nothing is placed.

5. Add a floor and a roof

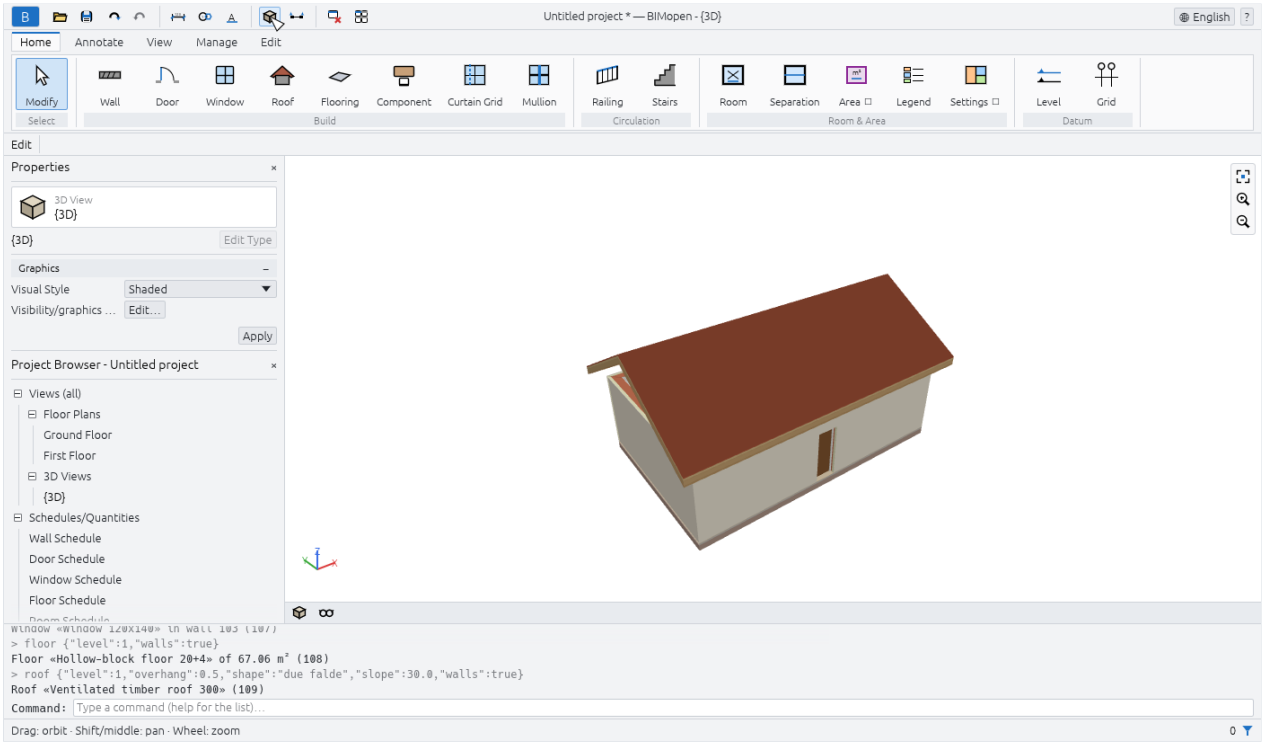
1. Click **Home ▸ Build ▸ Flooring**. The contextual tab **Edit | Create Floor Boundary** has a **Draw** panel with two ways of making the floor: **Boundary Line**, point by point, or **Pick Walls**.
2. Click **Pick Walls**: a floor is created on the outer perimeter of the walls of the level, and the command line gives its area.
3. Click **Home ▸ Build ▸ Roof**. On the options bar choose the shape (**Gable**, **Hip** or **Flat**), the **Slope** (30° by default) and the **Overhang** (500 mm by default).
4. Click **Pick Walls** in the **Draw** panel: the roof covers the walls of the level, with the overhang all around.

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6. Look at it in 3D

Click the **3D View** button (the cube) on the quick access toolbar. The default 3D view, **{3D}**, opens. Drag with the left mouse button to orbit, drag with the middle button (or with `Shift` and the left button) to pan, and turn the wheel to zoom. `Home` fits the whole model in the view.

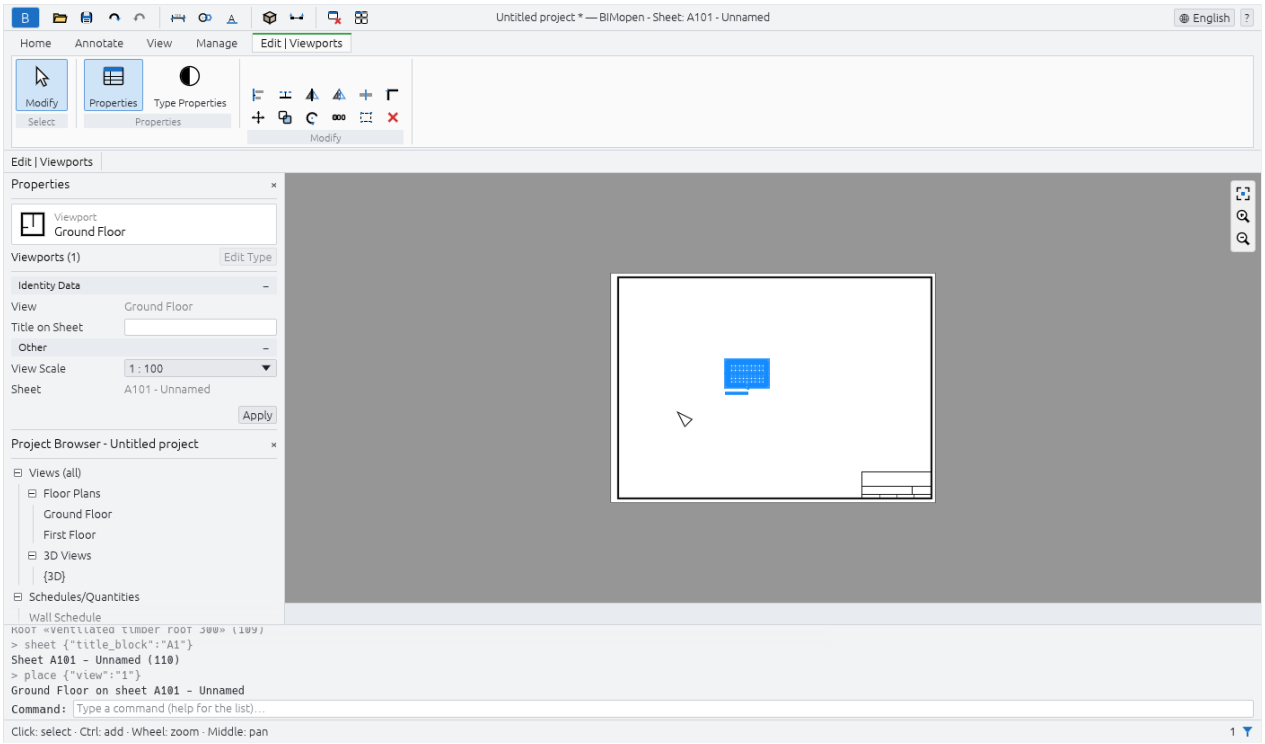


7. Put the plan on a sheet

1. Click the **View** tab, then **Sheet Composition ▸ Sheet**. The **New Sheet** dialog lists the title blocks: none, A0, A1, A2 or A3, with their size in millimetres.
2. Click **OK**. The sheet is created with the next free number (**A101** for the first one, then **A102**...) and opens.
3. Click **Sheet Composition ▸ View**. The **Views** dialog lists the views that are not on a sheet yet.

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4. Choose **Floor Plan: Ground Floor** and click **Add View to Sheet**. The plan is placed on the sheet at its own scale, 1 : 100, as a viewport, which stays selected.



8. Undo, redo and save

- `Ctrl+Z` undoes the last change (here, the viewport goes away); `Ctrl+Y` redoes it. The arrows of the quick access toolbar do the same.
- `Ctrl+S` saves the project. A project that has never been saved asks for a file name first; the file has the extension `.bimopen`. In the browser, the file is downloaded.

The title bar shows an asterisk after the project's name while there are unsaved changes: **Untitled project ***.

What next

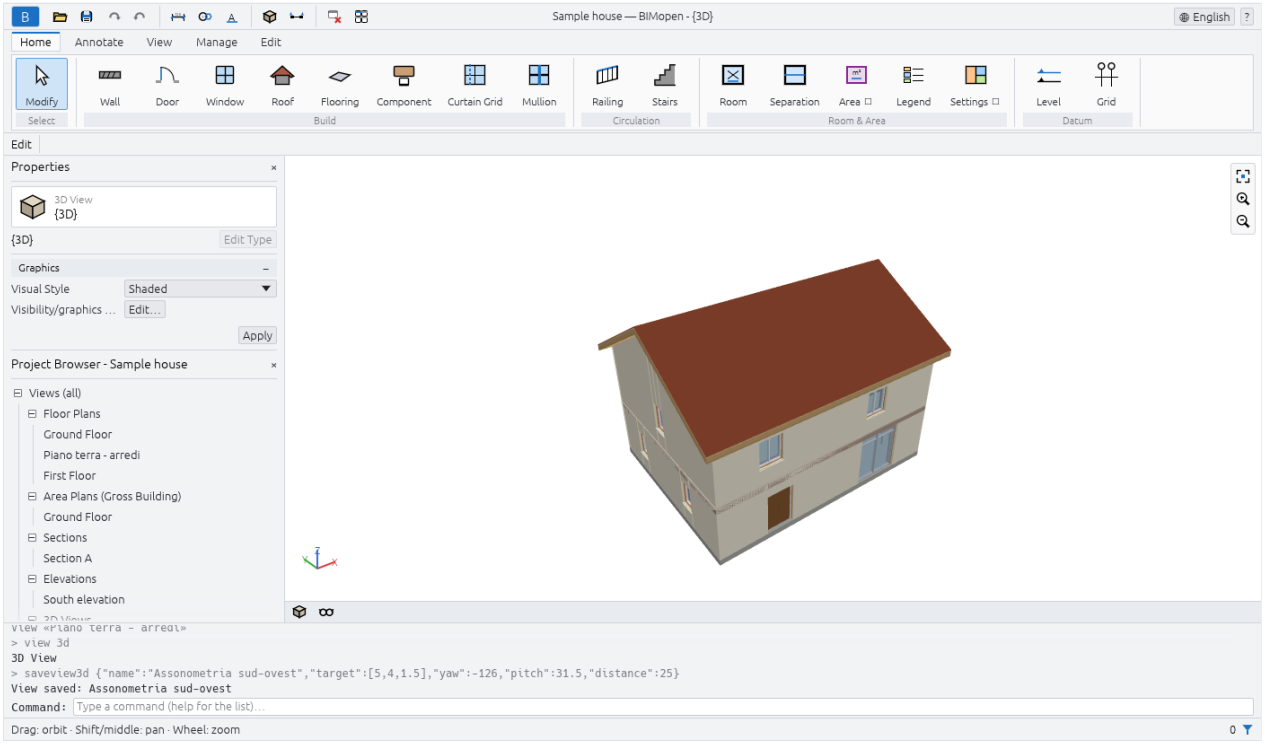
- The interface describes every part of the window.
- Keyboard shortcuts lists the keys and the two-letter shortcuts, such as `WA` for a wall.
- Command line and MCP shows how to type commands, run scripts and let an agent model for you.

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The interface

The BIMopen window has the same parts in the desktop and in the web version. From top to bottom:

1. the **title bar**, with the application menu, the quick access toolbar, the name of the project and of the active view, the language menu and the help menu;
2. the **ribbon**, with its tabs and panels of tools;
3. the **options bar**, with the settings of the active tool;
4. on the left, the **Properties** palette above the **Project Browser**;
5. the **drawing area**, where the views open in windows, each with its view control bar;
6. the **command line**, with the log of what has been done;
7. the **status bar**.

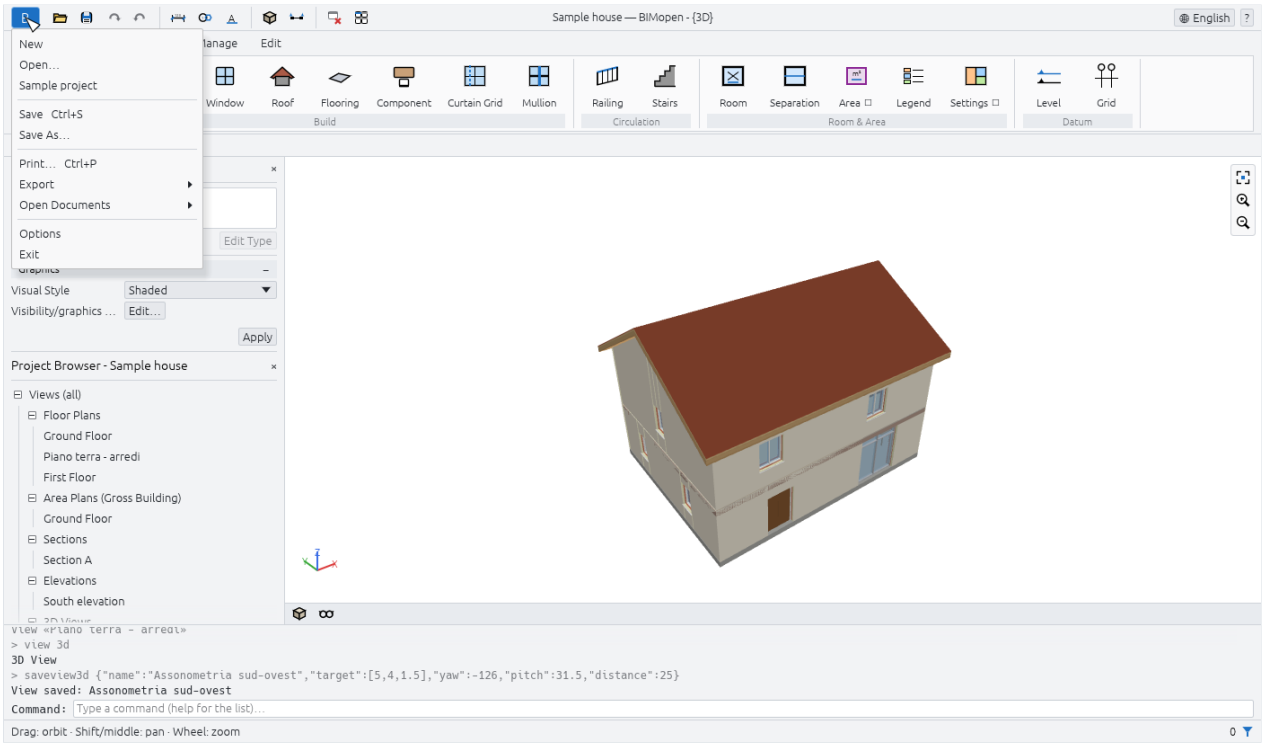


In Arabic the whole window is mirrored: the palettes are on the right, the ribbon and the options bar read from right to left, and the navigation bar is on the left edge of the views.

Application menu

The blue **B** button at the left end of the title bar opens the application menu: the commands that work on the whole project.

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Item	What it does
New	Closes the current project and starts a new one from the default template: two levels with their plans, and the default types
Open…	Opens a `.bimopen` project, or imports an `.rvt` model, chosen in a file dialog. In the browser, drag the file into the page instead
Sample project	Opens the sample house
Save `Ctrl+S`	Saves the project in its file; a project never saved asks for a name, like **Save As…**
Save As…	Saves the project in a new `.bimopen` file and goes on working on that file
Print… `Ctrl+P`	Prints sheets to PDF
Export ▸ Export IFC4…	Writes the model as an IFC4 file
Export ▸ Export plan SVG…	Writes the active plan as an SVG drawing
Export ▸ Export view PNG…	Writes the active view as a PNG image
Open Documents	Lists the open view windows; click one to bring it to the front
Options	Opens the **Options** dialog
Exit	Closes BIMopen (desktop only)

New, **Open…** and **Sample project** replace the current project without asking: save it first if you want to keep it.

> Warning
>
> Unsaved changes are lost when you choose **New**, **Open…** or **Sample project**. The asterisk after the project's name in the title bar tells you that there are unsaved changes.

Options

Application menu ▸ Options has two groups:

Setting	Values	Effect
User Interface ▸ Language	Italiano, English, Deutsch, Français, Español, العربية	The language of the whole interface, at once

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Setting	Values	Effect
User Interface ▸ Theme	**Light**, **Dark**	The colours of the window. **Dark** also turns on **Invert background color**
Graphics ▸ Invert background color	on, off	Views with a dark background and light lines

BIMopen remembers the language and the theme the next time it starts.

Quick access toolbar

Next to the application menu, the quick access toolbar keeps the most used commands one click away. Rest the mouse on a button to read what it does and its shortcut.

Button	What it does
Open… (folder)	As in the application menu
Save (disk)	Saves, `Ctrl+S`
Undo (arrow left)	Undoes the last change, `Ctrl+Z`. Greyed when there is nothing to undo
Redo (arrow right)	Redoes the change just undone, `Ctrl+Y`. Greyed when there is nothing to redo
Aligned dimension	Starts the aligned dimension tool, `DI`
Tag All	Tags the doors and windows of the open plan that have no tag yet, `TG`
Text	Starts the text tool, `TX`
3D View (cube)	Opens the default 3D view, **{3D}**
Section	Starts the section tool, `SE`
Close Hidden	Closes every view window except the active one. Greyed when only one is open
Switch Windows	Lists the open view windows; the active one is ticked

In the middle of the title bar, when the window is wide enough, BIMopen shows the name of the project and of the active view, for example **Sample house — BIMopen - Floor Plan: Ground Floor**. An asterisk after the project's name means that it has unsaved changes.

At the right end:

- the **language menu**, a globe with the name of the current language: choose another one and the interface changes at once;
- the **?** menu: **Commands** lists every command at the command line (as typing `help`), and **About** shows the version, the licence and the credits.

? ▸ Help (`F1`) opens this manual, in the language of the interface, in the browser. `F1` opens the page of what you are doing: the open dialog, the ribbon button under the mouse (its tip ends with **Press F1 for more information**) or the active tool.

Ribbon

The ribbon groups the tools in tabs, and each tab in panels. The name of a panel is written under its buttons. A button whose name is followed by a small mark opens a menu. Resting the mouse on a button shows what it does and, in brackets, the shortcut you can type to start it.

Tab	Panels
Home	**Select**, **Build**, **Circulation**, **Room & Area**, **Datum**
Annotate	**Select**, **Dimension**, **Text**, **Detail**, **Tag**
View	**Create**, **Graphics**, **Sheet Composition**, **Windows**
Manage	**Settings**, **Inquiry**
Edit	The contextual tab: its panels depend on the active tool and on the selection

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The Home tab

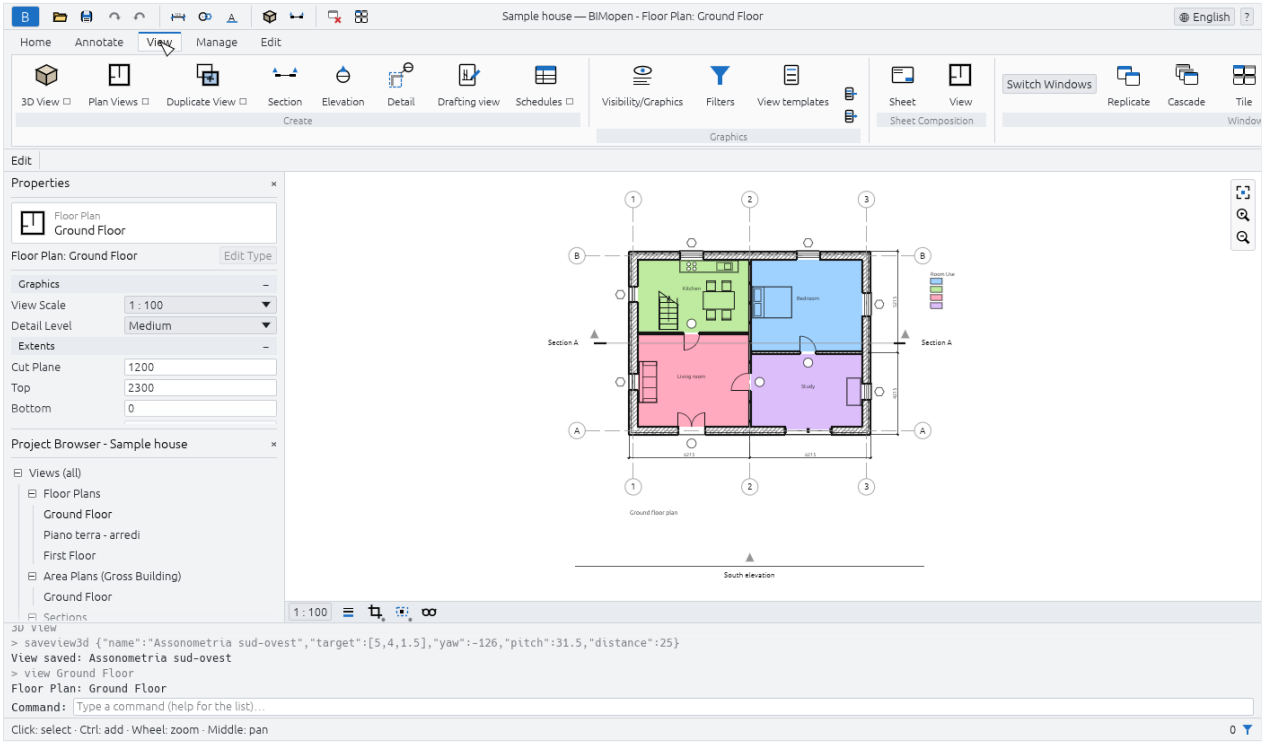
Panel	Buttons
Select	**Modify**: ends the active tool and goes back to selecting, `Esc`
Build	**Wall** `WA`, **Door** `DR`, **Window** `WN`, **Roof** `RF`, **Flooring** `SB`, **Component** `CM`, **Curtain Grid** `CG`, **Mullion** `MU`
Circulation	**Railing** `RA`, **Stairs** `ST`
Room & Area	**Room** `RM`, **Separation** `RS`, **Area** ▾ (**Area Plan**, **Area Boundary Line**, **Area**), **Legend** `LG`, **Settings** ▾ (**Area and Volume Computations**, **Color Schemes**)
Datum	**Level**: a new level above the last one; **Grid** `GR`

Area Boundary Line and **Area** work in an area plan: until one is open they are greyed, and their tip says to open an area plan first.

The Annotate tab

Panel	Buttons
Select	**Modify**
Dimension	**Aligned** `DI`, **Linear** `DL`, **Angular** `DA`
Text	**Text** `TX`
Detail	**Detail line**, **Filled region**, **Masking region**, **Detail component** `CD`, **Repeating detail** `RD`, **Insulation** `IS`
Tag	**Tag All** `TG`

The View tab



Panel	Buttons
Create	**3D View** ▾ (**Default 3D View**, **Save View**), **Plan Views** ▾ (**Floor Plan**, **Area Plan**), **Duplicate View** ▾ (**Duplicate**, **Duplicate with Detailing**), **Section** `SE`, **Elevation** `EL`, **Detail** `DT` (a callout), **Drafting view** `VD`, **Schedules** ▾ (**Schedule/Quantities**, **Predefined schedules**)
Graphics	**Visibility/Graphics** `VG`, **Filters**, **View templates**, and two small buttons: **Apply template to current view**, **Create template from current view**
Sheet Composition	**Sheet**: a new sheet with the title block you choose; **View**: places a view on the open sheet

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Panel	Buttons
Windows	**Switch Windows**, **Replicate**, **Cascade**, **Tile**, **Maximize**, **Close Hidden**, **User Interface**

Save View works only while a 3D view is open, **Duplicate with Detailing** only for views that are not 3D, and **Sheet Composition ▸ View** only while a sheet is open.

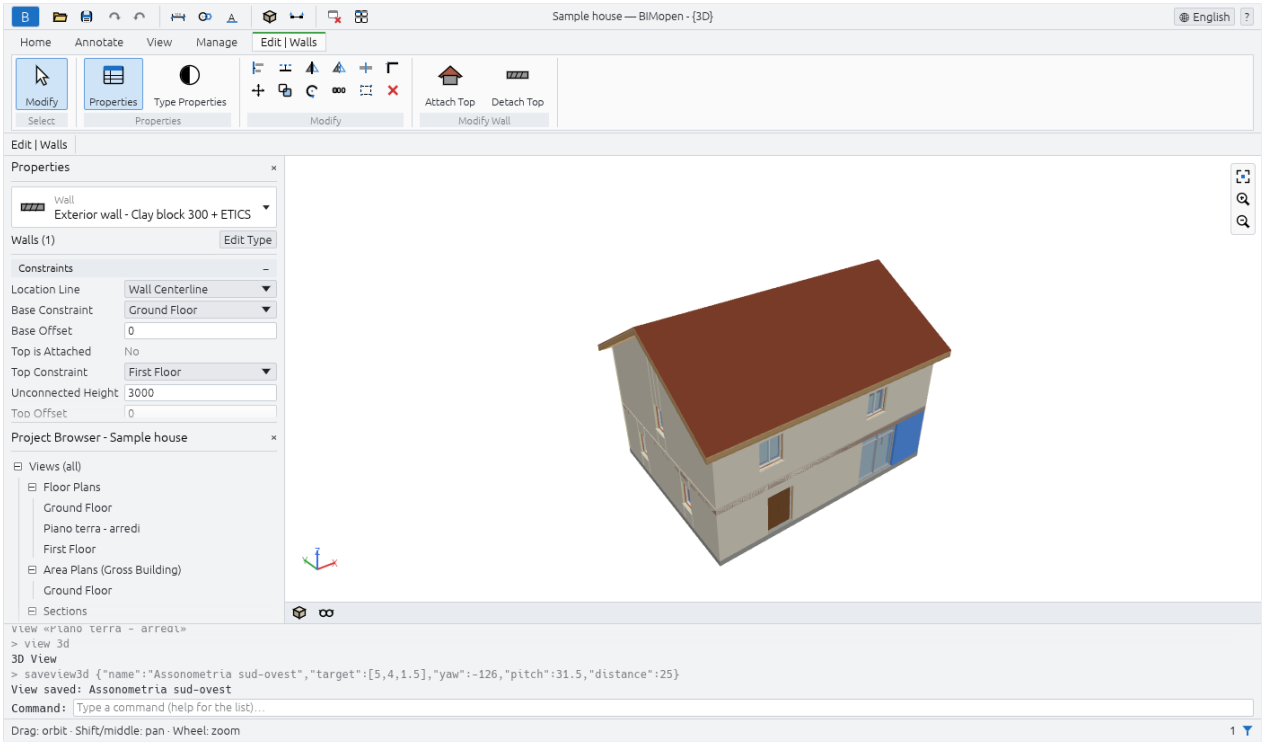
The Manage tab

Panel	Buttons
Settings	**Materials**, **Project information**, **Object styles**, **Line styles**, **Purge unused**, **Project units** `UN`, **Snap**, and **Additional settings** (**Line styles**, **Fill patterns**)
Inquiry	**Check**: checks the openings and references of the model and reports at the command line

Snap turns snapping on and off for the whole program: it is highlighted while on.

The contextual tab

When you start a tool or select something, the ribbon switches by itself to the last tab, which takes the name of what you are doing: **Edit | Place Wall** while drawing walls, **Edit | Walls** with walls selected, **Edit | Multi-Select** with elements of different categories. The tab is marked with its own colour. When the tool ends, or the selection is cleared, the ribbon goes back to the tab you were using.



It always has these panels:

Panel	Buttons
Select	**Modify**
Properties	**Properties**: shows or hides the Properties palette; **Type Properties**: the properties of the type of the selection, or of the next element the tool will place
Modify	Twelve small tools in two rows: **Align** `AL`, **Offset** `OF`, **Mirror - Pick Axis**, **Mirror - Draw Axis** `MM`, **Split Element** `SL`, **Trim/Extend to Corner** `TR`; **Move** `MV`, **Copy** `CO`, **Rotate** `RO`, **Array** `AR`, **All** (`Ctrl+A`), **Delete** (`Del`)

Delete works only when something is selected. The selecting and modifying chapter describes the Modify tools.

Other panels appear for what is selected or for the tool:

When	Panel and buttons
Doors or windows selected	**Opening**: **Flip facing**, **Flip hand**

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When	Panel and buttons
Walls selected	**Modify Wall**: **Attach Top** (the walls rise to the roof above them), **Detach Top**
A level selected	**View**: **Open plan**
Sections or elevations selected	**Section**: **Split Segment** (sections only), **Flip**, **Go to View**
A callout selected	**Detail**: **Go to View**
Curtain walls selected	**Curtain Wall**: **Curtain Grid**, **Mullion**, **Panel**
Flooring or **Roof** tool	**Draw**: **Boundary Line**, **Pick Walls**
Stairs or **Railing** tool	**Mode**: **Finish Edit Mode**; **Draw**: **Run** or **Lines**
Mullion tool	**Placement**: **Grid Line**, **Grid Line Segment**, **All Grid Lines**
A detail tool	The drawing shapes of the tool

Options bar

Under the ribbon, the options bar starts with the name of the contextual tab, then shows the settings of the active tool. They apply to the next elements you place. At its end, while a tool is active, it shows the level you are working on: **Level: Ground Floor**.

Tool	Settings on the options bar
Wall	**Height** or **Depth**; the level the wall reaches, or **Unconnected** with its value; **Location Line**; **Chain**; **Offset**; **Radius** (greyed: curved walls are not available yet); **Exterior on the left** or **Exterior on the right** (the space bar swaps them)
Roof	The shape: **Gable**, **Hip** or **Flat**; **Slope**, from 5° to 60°; **Overhang**, up to 2 m
Stairs	**Width**, in millimetres; **Railing Type**, or **None**
Dimensions	What the cursor picks first on a wall: **Wall centerlines** or **Wall faces**
Detail (callout)	The scale of the new detail view
Area Boundary Line	**Line** or **Pick lines**; with **Pick lines**, **Apply area rules** or an **Offset**
Tools that draw lines	**Ortho (Shift)**: horizontal and vertical segments only; **Snap**: snapping to endpoints, midpoints and intersections

With **Ortho (Shift)** off, holding `Shift` while drawing gives the same constraint; with it on, `Shift` releases it.

Properties

The **Properties** palette shows and changes the parameters of what you are working on. From top to bottom it has:

- the **type selector**: the family in small letters and the type underneath. Click it to choose another type: the selected elements change type, or, while a tool is active, the next element gets that type;
- a row with what the palette is showing, for example **Walls (1)** or **New Wall**, and the **Edit Type** button, which opens the type properties;
- the parameters, in groups (**Constraints**, **Graphics**, **Identity Data**...). Click the name of a group to fold it. Values in grey are read-only: they are computed from the model;
- the **Apply** button, which confirms a value you are typing. `Enter` does the same.

What the palette shows depends on the situation:

Situation	The palette shows
Nothing selected, no tool	The properties of the active view (scale, detail level, visual style...), or of the view clicked in the Project Browser
A tool active	The type of the next element, as **New Wall**, **New Door**...
One element selected	Its type and its instance parameters
Several elements of one category	Their number, as **Walls (3)**

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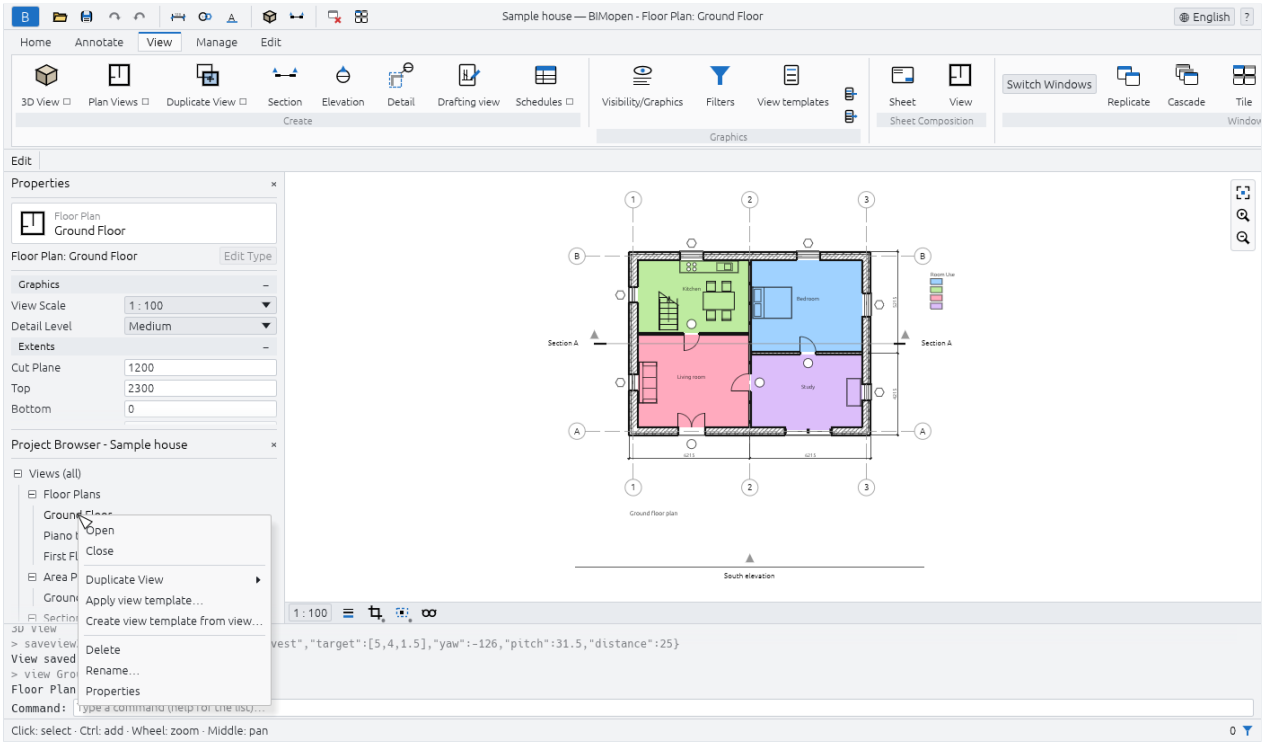
Situation	The palette shows
Elements of different categories	**Multi-Select**, **Common (5)**, and how many of each category, as **3 × Walls**

Lengths are written and read in the project's units (millimetres in a new project). You can type a value with its unit, for example `2.7m`.

To close the palette, click × on its title; to show it again, use **View ▸ Windows ▸ User Interface ▸ Properties** or the **Properties** button of the contextual tab.

Project Browser

The **Project Browser** lists the views, schedules, sheets and families of the project in a tree. Its title carries the name of the project. Click ⊟ or ⊞ to fold or unfold a branch.



Branch	What it contains
Views (all) ▸ Floor Plans	The floor plans of every level, from the lowest, and the callouts drawn in plans
Views (all) ▸ Area Plans (...)	The area plans, one branch per area scheme
Views (all) ▸ Sections, Elevations	The sections and the elevations
Views (all) ▸ Detail Views	The callouts drawn in sections
Views (all) ▸ 3D Views	**{3D}**, the default 3D view, and the saved 3D views
Views (all) ▸ Drafting views	The drafting views
Schedules/Quantities	The project's schedules, then the predefined schedules
Sheets (all)	The sheets, as number - name
Families	The types of the project, by category: walls, doors, windows, floors, roofs, components and the others

The branches of area plans, sections, elevations, detail views and drafting views appear only when they have something in them. The active view is written in bold.

With a view:

- a **click** shows its properties in the Properties palette, without opening it;

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- a **double-click** opens it;
- a **right-click** opens its menu: **Open**, **Close** (closes its windows), **Duplicate View ▸** (**Duplicate**, **Duplicate with Detailing**), **Apply view template...**, **Create view template from view...**, **Delete**, **Rename...**, **Properties**.

The menu of a sheet has **Apply view template to all views...** instead of the two template items, and no **Duplicate View**. **{3D}** cannot be deleted or renamed.

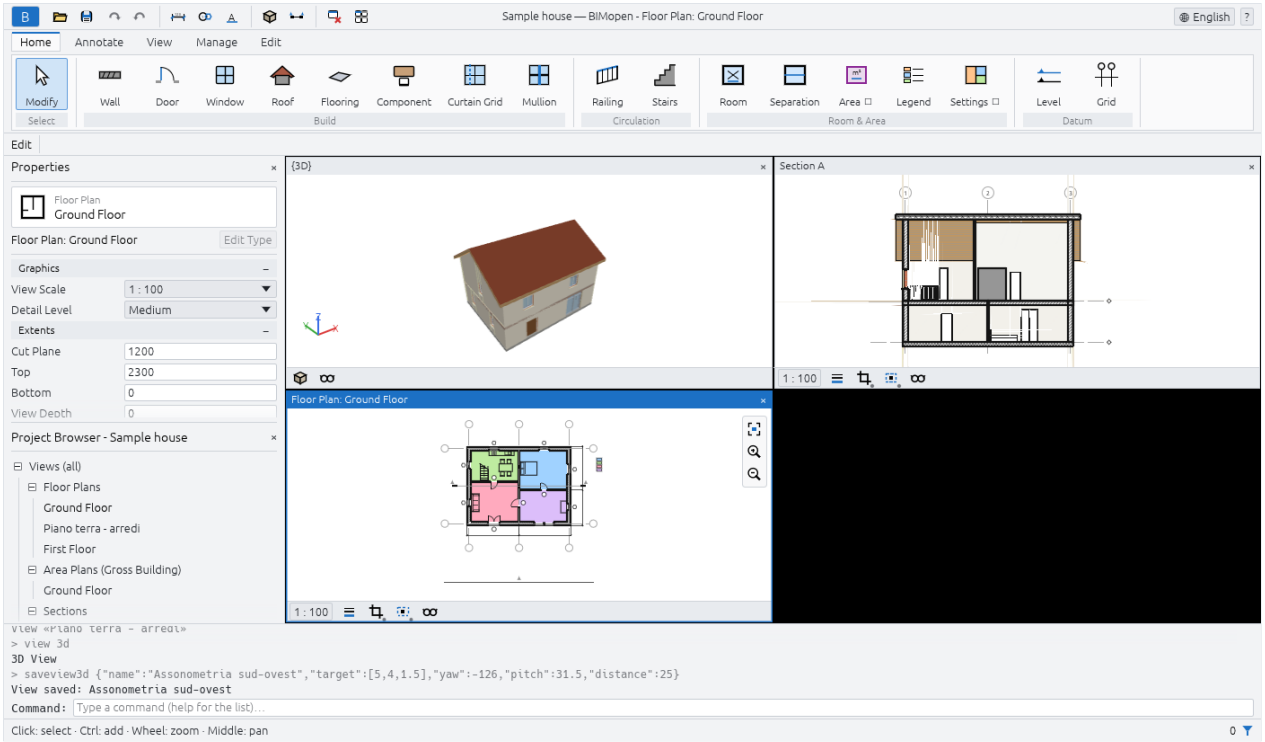
With a schedule of the project, a click selects it, a double-click opens it, and the right-click menu has **Open**, **Schedule Properties...** and **Delete**. A predefined schedule opens with a click.

With a type, under **Families**, a double-click opens its type properties; the right-click menu has **Type Properties**, **Duplicate...**, **Create Instance** (starts the tool that places it, with that type) and **Delete**.

Drawing area and view windows

Every view you open is a window in the drawing area. A view opened again comes to the front; at least one window always stays open. **View ▸ Windows** arranges them:

Button	What it does
Switch Windows	Lists the open windows; the active one is ticked
Replicate	Opens a second window on the active view, with its own zoom
Cascade	Overlapping windows, each with its title bar: drag a title to move a window, its edges to resize it
Tile	All the windows side by side, each with a title bar; the active one has a blue title and frame
Maximize	The active window fills the drawing area; the others wait underneath
Close Hidden	Closes every window except the active one
User Interface	Shows or hides the **Project Browser**, **Properties**, the **Command line**, the **Status bar** and the **Navigation bar**



Click in a window to make it active: the tools work in the active window, the others only show the model. In **Tile** and **Cascade**, the **×** on a window's title bar closes it.

View control bar

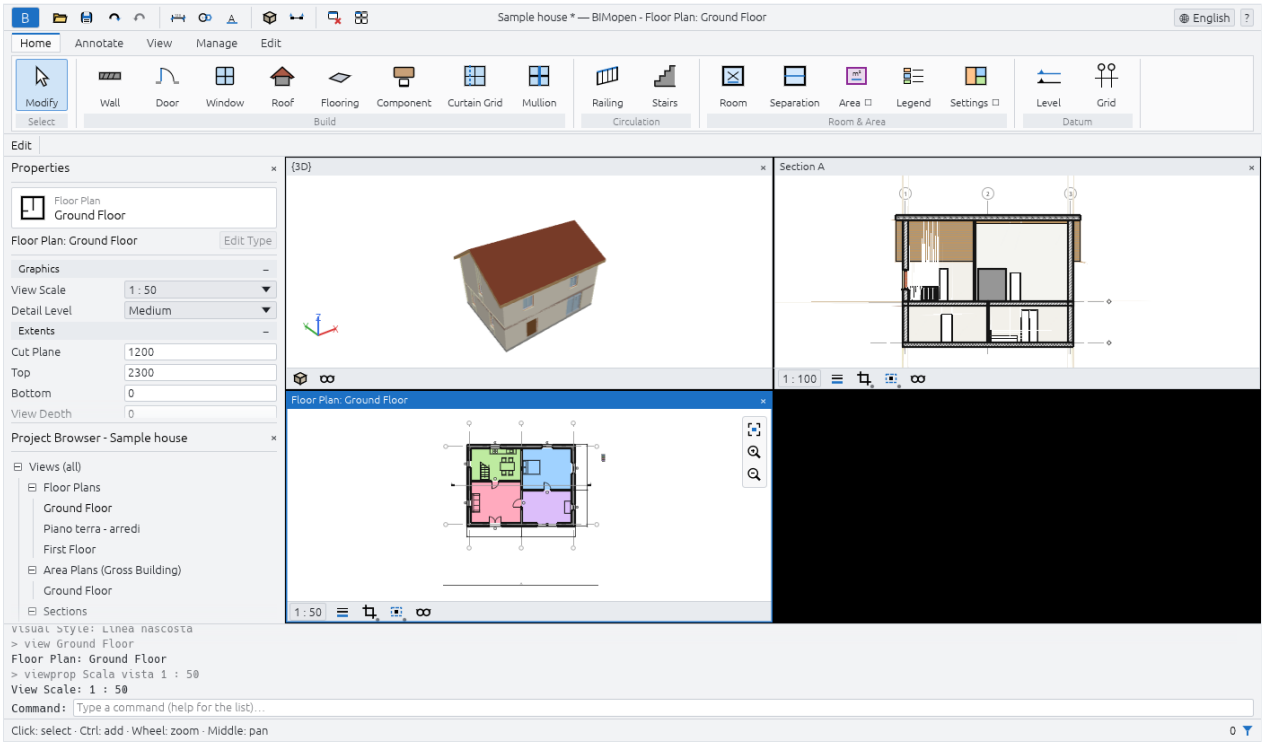
At the bottom of every plan, section, elevation, callout and drafting view:

Control	What it does
The scale, for example **1 : 100**	Opens the list of scales; under **Custom** type another one, as `1 : 75`, and press `Enter`

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Control	What it does
Detail level	**Coarse**, **Medium** or **Fine**
Crop View	Crops the view to its crop region, or stops cropping; a blue dot shows it is on
Show Crop Region	Shows or hides the crop region's frame; a blue dot shows it is on
Temporary Hide/Isolate	Hides or isolates elements or categories in this view until you reset it; framed while active

In a 3D view, the bar has the **Visual style** (**Wireframe**, **Hidden Line**, **Shaded**, **Shaded with Edges**, **Consistent Colors**) and **Temporary Hide/Isolate**. A sheet has no view control bar: its views have their own scale.



Navigation bar

On the right edge of the active view (the left edge in Arabic), three buttons: **Zoom to Fit** (the same as `Home`), **Zoom In** and **Zoom Out**.

Moving around in the views

In plans, sections, elevations and sheets	
Mouse wheel	Zooms in and out at the cursor
Drag with the middle button	Pans
`Shift` + drag with the left button	Pans, while no tool is active
`Home`	Fits the whole view in the window

In 3D views	
Drag with the left (or right) button	Orbits around the model
Drag with the middle button, or `Shift` + left button	Pans
Mouse wheel	Zooms in and out
`Home`	Fits the whole model in the window

A small tripod in the lower corner of the 3D view shows the directions of the axes: X red, Y green, Z blue. A saved 3D view remembers where you left it.

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The context menu of a view

Right-click in a view while no tool is active. If the cursor is on an element that is not selected, the element is selected first. The menu has:

- **Cancel**: clears the selection;
- **Hide in view ▸ Elements** or **▸ Category**: hides the selected elements, or their whole categories, in this view;
- **Override graphics in view ▸ By category...**: opens **Visibility/Graphics** on the categories of the selection;
- **Visibility/Graphics...**.

While a tool is active, a right-click does what `Esc` does: it cancels the step in progress.

Command line

Under the drawing area, the command line shows the log of what has been done, the most recent at the bottom: the commands run, in grey after a `>`, their results, and errors in red. On its last row, the prompt says what the active tool expects, for example `WALL next point [Esc to finish]:`, or `Command:` when no tool is active.

You do not need to click it: whatever you type while no field is being edited goes to the command line. `Enter` runs the line; `Up` and `Down` bring back the lines typed before. The command line chapter describes the commands.

Status bar

On the left, the status bar shows:

- while a tool is active, what the tool expects (the same prompt as the command line);
- with no tool, the category and the type of the element under the cursor, as **Walls : Exterior wall - Clay block 300 + ETICS**;
- otherwise a reminder of the mouse: **Click: select · Ctrl: add · Wheel: zoom · Middle: pan** in plans, **Drag: orbit · Shift/middle: pan · Wheel: zoom** in 3D.

On the right, a funnel and the number of selected elements; in a plan, also the coordinates of the cursor in metres.

Selecting, in short

The selecting and modifying chapter will describe selection in full. The essentials:

- click an element to select it; click an empty point to clear the selection;
- `Ctrl` + click adds an element to the selection, or removes it if it is already selected;
- drag from left to right to select what is entirely inside the rectangle, from right to left to select what it touches;
- drag a selected element to move it;
- `Ctrl+A` selects the elements of the open plan's level, or of the whole model in a 3D view; `Del` deletes the selection; `Esc` clears it.

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Projects and files

This chapter will describe how a project is created, opened, saved and exchanged with other programs.

> Being written
>
> This chapter is completed with area R7 of BIMopen's realignment, *Documentation and files*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- **New**, **Open…**, **Save**, **Save As…** and the sample project, on the desktop and in the browser;
- the `.bimopen` file: what it contains and how it keeps working with newer versions of BIMopen;
- backups and recovering unsaved work;
- **Project information**: name, number, client and address;
- **Project units**: lengths, areas, volumes, angles and slopes, with their rounding and symbols;
- exporting to IFC4, SVG, PNG and CSV, with their options;
- importing `.rvt` models: what is read and what is not.

Meanwhile

Getting started shows how to save, and Command line and MCP lists the export formats of `bimopen-cli convert`. At the command line, `help` followed by the name of a command shows its usage.

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Selecting and modifying

This chapter will describe how elements are selected and changed.

> Being written
>
> This chapter is completed with area R2 of BIMopen's realignment, *Selection and modify*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- highlighting under the cursor, `Tab` to cycle between overlapping elements and chains of walls;
- selection windows from left to right and from right to left, `Ctrl` and `Shift`;
- filtering a selection and **Select All Instances**;
- dragging elements in every view: joined walls stretch, hosted elements follow;
- the arrow keys, temporary dimensions, pinning;
- the tools of the **Modify** panel: **Move**, **Copy**, **Rotate**, **Mirror**, **Array**, **Align**, **Offset**, **Trim/Extend**, **Split Element**, **Delete**;
- undo and redo.

Meanwhile

The interface summarises selection, and Keyboard shortcuts lists the keys and the typed shortcuts of the Modify tools. At the command line, `help` followed by the name of a command shows its usage.

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Walls, doors and windows

This chapter will describe the walls of the building and the openings they host.

> Being written
>
> This chapter is completed with area R3 of BIMopen's realignment, *Walls, doors, windows*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- the **Wall** tool: its options bar, the location line, chains, offsets and the exterior side;
- how walls join at corners, in T and in X, and how a join is prevented;
- wall types and their layers: **Edit Type**, **Duplicate**, the structure of the layers;
- the base and top of a wall: levels, offsets, **Attach Top** to roofs and floors;
- doors and windows: placement, host wall, flipping, sill and head heights, tags;
- how walls, doors and windows are drawn in plan at each detail level, in section and in 3D.

Meanwhile

Getting started draws four walls, a door and a window step by step. At the command line, `help` followed by the name of a command shows its usage.

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Floors, roofs, stairs and components

This chapter will describe the horizontal elements of the building, its stairs and the objects placed in it.

> Being written
>
> This chapter is completed with area R4 of BIMopen's realignment, *Floors, roofs, stairs, railings, curtain walls, components*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- sketch mode: boundaries drawn line by line or picked from walls, and **Finish Edit Mode**;
- floors and their types;
- roofs by footprint: shapes, slopes, overhangs, walls attached to the roof;
- stairs by run, landings, and the railings that follow them;
- railings drawn along a path;
- curtain walls: grids, mullions and panels;
- components from the library: furniture and plumbing fixtures.

Meanwhile

Getting started makes a floor and a roof from the walls. At the command line, `help` followed by the name of a command shows its usage.

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Views and graphics

This chapter will describe the views of the model and how they are drawn.

> Being written
>
> This chapter is completed with area R5 of BIMopen's realignment, *Views and graphics*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- floor plans and their view range;
- sections, elevations and callouts: their lines, extents and segments;
- 3D views and their visual styles, saved 3D views;
- crop regions, temporary hide and isolate, hiding in view;
- **Visibility/Graphics**, **Object styles**, line styles and line weights;
- view filters and view templates;
- materials and fill patterns, in plan, in section and in 3D.

Meanwhile

The interface describes the view windows, the view control bar and how to move around in the views. At the command line, `help` followed by the name of a command shows its usage.

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Annotation and 2D details

This chapter will describe dimensions, text, tags, rooms, areas and the 2D drawing tools.

> Being written
>
> This chapter is completed with area R6 of BIMopen's realignment, *Annotation and 2D*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- aligned, linear and angular dimensions, their constraints (lock and EQ) and temporary dimensions;
- text notes and tags;
- rooms, room separation lines and room tags;
- area plans, area schemes, area boundaries and areas;
- color schemes and their legends;
- detail lines, filled and masking regions, detail components, repeating details and insulation;
- drafting views.

Meanwhile

Keyboard shortcuts lists the typed shortcuts of the annotation tools. At the command line, `help` followed by the name of a command shows its usage.

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Sheets, schedules and printing

This chapter will describe how the project is documented: sheets, schedules and printed drawings.

> Being written
>
> This chapter is completed with area R7 of BIMopen's realignment, *Documentation and files*: each feature is compared with the reference behaviour, corrected where needed and checked through the interface, and only then described here. Until then, this page lists what it will cover.

What this chapter will cover

- sheets and title blocks;
- viewports: placing views on sheets, their scale and title;
- schedules as views of the model: fields, filters, sorting and grouping, totals, editing values in the schedule;
- the predefined schedules and the material takeoff;
- printing sheets to PDF;
- exporting schedules to CSV.

Meanwhile

Getting started places a plan on a sheet step by step. At the command line, `help` followed by the name of a command shows its usage.

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Command line and MCP

Everything you do in BIMopen is a command. A click on **Wall** and two points in the plan run the same `wall` command you can type yourself. This chapter describes the command line of the window, the `bimopen-cli` program, which runs the same commands without a window, and the MCP server, through which an AI agent can model, check and render a project.

The command line in the window

The command line sits under the drawing area. Its upper part is the log; its last row is where you type.

- You do not need to click it first: whatever you type while no other field is being edited goes there.
- `Enter` runs the line. The command is echoed in the log after a `>`, followed by its result; an error is shown in red and changes nothing.
- `Up` and `Down` bring back the lines typed before.
- On the left of the row, the prompt says what is expected: `Command:` when no tool is active, or the request of the active tool, such as `WALL next point [Esc to finish]:`.
- `help` lists every command with a line of description; `help wall` shows the usage of one. **?** ▸ **Commands** in the title bar does the same as `help`.

Typed shortcuts start the tools

A command typed **without arguments** starts the interactive tool, as if you clicked its button: `WA` (or `wall`) starts the wall tool, `DR` the door tool, `VG` opens Visibility/Graphics. The same command **with arguments** does the work at once, without the tool: `wall 0,0 5,0` draws a wall from the origin to 5 m along x.

Command names, their short forms and their arguments are the same in every language of the interface; most also accept an Italian name, such as `muro` for `wall`. Parameter names, type names and view names follow the language of the interface: in English you type `set Unconnected Height 2.7m`, in Italian `set Altezza non connessa 2.7m`. The keyboard shortcuts chapter lists every short form.

Points and lengths

You type	It means
`3,4`	The point x = 3 m, y = 4 m
`@2,0`	2 m along x and 0 along y from the previous point
`@4<90`	4 m from the previous point, at 90° (angles counter-clockwise from the x axis)
`2.7m`, `270cm`, `2700mm`	A length with its unit
`2.7` or `2,7`	A length in metres: at the command line a number without unit is in metres

For example, `wall 0,0 @6,0 @4<90 0,4 c` draws a 6 × 4 m rectangle; the final `c` closes it.

> Warning
>
> At the command line `set Unconnected Height 2700` means 2700 **metres**. Write the unit: `2700mm`. In the Properties palette, instead, a number without unit is in the project's units, millimetres in a new project.

Arguments in JSON

Every command also takes its arguments as a JSON object, which gives access to all its options. The usage line of `help` lists them:

```
wall 0,0 10,0 {"height":3.2,"location":"Linea d'asse del muro"}
roof walls {"slope":30,"overhang":0.5}
sheet A102 Plans {"title_block":"A3"}
export doors.csv {"schedule":"doors"}
```

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Commands you will use often

Command	What it does
`new`, `sample`	A new project, the sample project
`open house.bimopen`, `save house.bimopen`	Opens and saves a project
`undo`, `redo`	Undoes and redoes the last change
`view 3d`, `view Ground Floor`	Opens the default 3D view, or the plan of a level
`wall`, `door`, `window`, `floor`, `roof`	Model elements: see the usage with `help`
`select walls`, `select all`, `deselect`	Selects by category, everything, nothing
`set <parameter> <value>`	Sets a parameter of the selection, as in the Properties palette
`type <type name>`	Changes the type of the selection, or of the next element
`move 1,0`, `copy 0,3`, `rotate 90`	Modifies the selection
`delete`	Deletes the selection
`list walls`, `params 102`, `info`	Lists elements, the parameters of one, a summary of the project
`schedule doors`	Shows a schedule at the command line
`export house.ifc`	Exports IFC, SVG, PNG or CSV, by the file's extension

Every element has a number, its id, shown in brackets after it is created: `Door «Single swing door 80x210» in wall 102 (106)`. Commands accept it where they need an element: `params 106`, `delete 106`.

bimopen-cli

`bimopen-cli` runs BIMopen without a window: to convert files, to build a project from a script, to check a model, or to serve agents over MCP. Build it with `cargo build --release -p bimopen-cli`.

```
bimopen-cli info <file>
bimopen-cli convert <in> <out> [--level NAME] [--view 3d] [--schedule walls|doors|windows|floors|rooms|materials]
bimopen-cli run [<file> | --sample] [--cmd LINE]… [--script FILE] [--save FILE] [--export FILE]…
bimopen-cli commands
bimopen-cli mcp [<file> | --sample]
```

Subcommand	What it does
`info <file>`	A summary of the project and, in JSON, its levels, element counts and active view
`convert <in> <out>`	Opens `<in>` (`.bimopen` or `.rvt`) and writes `<out>`: `.ifc`, `.svg` or `.png` of a plan (the lowest level, or `--level`), `.png` of the 3D view (`--view 3d`), `.csv` of a schedule (`--schedule`), or `.bimopen`
`run`	Opens a file, the sample, or a new project, then runs the options in order: `--cmd` one line, `--script` a file of lines (empty lines and lines starting with `#` are skipped), `--save` and `--export` a file
`commands`	Every command, with its description and usage
`mcp`	The MCP server on standard input and output
`--version`	The version

The messages are in the language of the environment variable `BIMOPEN_LANG` (`it`, `en`, `de`, `fr`, `es`, `ar`), or of the system (`LANG`), or Italian. `bimopen-cli` ends with code 0 when everything worked, 1 when a command failed (the message says which), and 2 when the arguments are wrong (it prints the usage).

```
export BIMOPEN_LANG=en
bimopen-cli run --script house.txt --save house.bimopen --export house.ifc
bimopen-cli convert house.bimopen plan.svg
bimopen-cli convert house.bimopen doors.csv --schedule doors
bimopen-cli convert house.bimopen first.png --level "First Floor"
bimopen-cli convert house.bimopen house-3d.png --view 3d
bimopen-cli convert project.rvt project.bimopen
```

A script is a text file with one command per line, exactly as typed at the command line:

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```
# a small house
wall 0,0 10,0 10,6 0,6 c
door 5,0
window 10,3
floor walls
roof walls
```

The schedules are written as CSV with `;` between the columns, the headings in the language of the messages and a total row at the end.

MCP: BIMopen for AI agents

The Model Context Protocol (MCP) lets an AI agent use programs as tools. `bimopen-cli mcp` is an MCP server: it speaks JSON-RPC 2.0 on standard input and output, one message per line, and keeps one project open for the whole conversation. Without a file it starts from a new project; `--sample` starts from the sample house.

To add it to an agent, give the agent the command to start. For example, in Claude Code:

```
claude mcp add bimopen -- bimopen-cli mcp --sample
```

or, in a client configured with JSON:

```
{ "mcpServers": { "bimopen": { "command": "bimopen-cli", "args": ["mcp", "--sample"] } } }
```

The server offers seven tools:

Tool	Arguments	What it does
`command_line`	`line`	Types a line at the command line, exactly as a person would
`execute`	`command`, `args`	Runs a command with its arguments in JSON
`inspect_model`	–	A summary: levels, element counts, active view, selection
`query_elements`	`category` (optional)	The elements, with id, category and name: `walls`, `doors`, `windows`, `floors`, `rooms`, `roomseps`, `grids`, `levels`, `types`
`get_params`	`id`	Every parameter of an element
`schedule`	`kind`	A schedule: `walls`, `doors`, `windows`, `floors`, `rooms` or `materials`
`render`	`view`, `width`, `height`, `yaw`, `pitch`	A PNG image of a view: `3d`, the name of a level for its plan, of a section, elevation, callout or drafting view, or the number or name of a sheet; `yaw` and `pitch` turn the 3D camera, in degrees

The answers carry the message of the command and, when there is one, its data in JSON, such as the ids of the elements created. An agent can check its own work by asking for a render.

> Tip
>
> Ask the agent to render the plan and the 3D view after each step: it sees what it built and corrects itself.

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Keyboard shortcuts

BIMopen has two kinds of shortcuts: **keys**, such as `Ctrl+Z`, which act at once, and **typed shortcuts**, such as `WA`, which you type letter by letter and confirm with `Enter`, like any command. Typed shortcuts are the same in every language.

On macOS, `Cmd` takes the place of `Ctrl`.

Keys

Key	What it does
`Ctrl+Z`	Undoes the last change
`Ctrl+Y` or `Ctrl+Shift+Z`	Redoes the change just undone
`Ctrl+S`	Saves the project; a project never saved asks for a file name
`Ctrl+P`	Opens **Print**
`Ctrl+A`	Selects the elements of the open plan's level, or of the whole model in a 3D view
`Del`	Deletes the selected elements, and the elements they host
`Home`	Fits the whole view in its window
`F1`	Opens the page of this manual about the open dialog, the ribbon button under the mouse or the active tool
`Esc`	Cancels the step in progress; pressed again, ends the tool; with no tool, clears the selection
`Enter`	Runs the line typed at the command line; confirms a value in the Properties palette
`Up` / `Down`	At the command line, the lines typed before
`Space`	While drawing walls, swaps the exterior and interior side; while placing a component or a detail component, rotates it
`Shift`	While drawing, keeps segments horizontal or vertical (or frees them, when **Ortho** is on); with the left button, pans the view
`Ctrl` + click	Adds an element to the selection, or removes it

`Ctrl+Z`, `Ctrl+Y`, `Ctrl+A`, `Del` and `Home` act on the model only while you are not typing in a field: in a field they act on the text.

In a floor, roof, stair or railing sketch, `Esc` (or a double-click) also finishes the sketch when it has enough points.

In the **Text Note** dialog, `Ctrl+Enter` inserts the note and `Esc` cancels it.

Mouse

	In plans and sections	In 3D views
Wheel	Zoom at the cursor	Zoom
Middle button, drag	Pan	Pan
`Shift` + left button, drag	Pan	Pan
Left button, drag	Selection window (left to right: inside; right to left: touching), or moves the selected elements	Orbit
Right button	Context menu; with a tool active, cancels the step	Orbit (drag)
Double-click	Finishes a sketch	–

Typed shortcuts

Type the letters while no field is being edited, then press `Enter`. Without arguments, a shortcut starts the tool or opens the dialog; with arguments, it runs the command at once (see Command line and MCP).

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Modelling

Shortcut	Command	Starts
`WA` or `W`	`wall`	Wall
`DR`	`door`	Door
`WN`	`window`	Window
`SB`	`floor`	Floor (the **Flooring** button)
`RF`	`roof`	Roof
`CM`	`component`	Component
`ST`	`stair`	Stairs
`RA`	`railing`	Railing
`CG`	`curtaingrid`	Curtain grid
`MU`	`mullion`	Mullion
`CP`	`curtainpanel`	Curtain wall panel
`GR`	`grid`	Grid
`RM`	`room`	Room
`RS`	`roomsep`	Room separation
`PA`	`areaplan`	The **Area Plan** dialog
`CA`	`areaboundary`	Area boundary line
–	`area`	Area
`LG`	`legend`	Color fill legend
`SC`	`colorscheme`	The **Color Schemes** dialog

Modifying

Shortcut	Command	Starts
`MV` or `M`	`move`	Move
`CO`	`copy`	Copy
`RO`	`rotate`	Rotate
`MM`	`mirror`	Mirror - Draw Axis
`AR`	`array`	Array
`OF`	`offset`	Offset
`AL`	`align`	Align
`TR`	`trim`	Trim/Extend to Corner
`SL`	`split`	Split Element
`DEL`	`delete`	Deletes the selection
`U`	`undo`	Undoes the last change

Annotating and detailing

Shortcut	Command	Starts
`DI`	`dim`	Aligned dimension
`DL`	`lineardim`	Linear dimension
`DA`	`angulardim`	Angular dimension
`TX`	`text`	Text

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Shortcut	Command	Starts
`TG`	`tag`	Tags the untagged doors and windows of the level
–	`detailline`	Detail line
`RG`	`region`	Filled region
–	`mask`	Masking region
`CD`	`detailcomponent`	Detail component
`RD`	`repeatingdetail`	Repeating detail
`IS`	`insulation`	Insulation

> Note
>
> The tip of the **Detail line** button shows `DL`, but `DL` starts the linear dimension, as the tip of **Linear** says. To start the detail line by typing, use `detailline`.

Views and settings

Shortcut	Command	Starts or opens
`SE`	`section`	Section
`EL`	`elevation`	Elevation
`DT`	`callout`	Callout (the **Detail** button)
`DS`	`splitsegment`	Split Segment of a section
`VD`	`draftingview`	The **Drafting view** dialog
`PP`	`floorplan`	The **Floor Plan** dialog
`DV`	`duplicateview`	Duplicates the active view
`VG` or `VV`	`vg`	**Visibility/Graphics**
`SO`	`objectstyle`	**Object styles**
`MA`	`material`	**Materials**
`UN`	`units`	**Project units**
`PU`	`purge`	**Purge unused**
–	`filter`	**Filters**
–	`viewtemplate`	**View templates**
–	`applytemplate`	Apply a view template to the active view
–	`linestyle`	**Line styles**
–	`fillpattern`	**Fill patterns**
–	`saveview3d`	Saves the {3D} view with a name
–	`renameview`	Renames the active view

The commands with **–** have no short form, but their whole name works the same way. `help` lists every command and its short forms.

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Troubleshooting

This chapter collects situations that can surprise you, and what to do about them. It grows with the manual: each area of the realignment adds its own.

Starting BIMopen

The web app shows «BIMopen non si è avviato» and an error

The browser could not give BIMopen a graphics surface: the message ends with `webgl2 not available` or with a WebGPU error. BIMopen in the browser needs WebGPU or WebGL2.

- Turn on hardware acceleration in the settings of the browser, then reload the page.
- Try another browser, or a newer version of the same one.
- Add `?webgl` to the address to use WebGL2 even where WebGPU exists but does not work.

The desktop app does not build on Linux

The interface needs the development files of xkbcommon, Wayland and GTK 3. On Debian and Ubuntu:

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev
```

The interface

The Properties palette, the Project Browser or the command line has disappeared

They are closed with the × on their title, or from **View ▸ Windows ▸ User Interface**. Tick them again in **User Interface**. The **Properties** button of the contextual tab also shows the palette again.

What I type ends up at the command line

Whatever you type while no field is being edited goes to the command line: this is how typed shortcuts work. To type in a field of the Properties palette or of a dialog, click the field first.

A button is greyed

Some tools work only in the right view, and their tip says why. **Area Boundary Line** and **Area** need an open area plan: the tip says **Open an area plan first**. **Save View** needs an open 3D view; **Sheet Composition ▸ View** needs an open sheet; **Undo** and **Redo** are greyed when there is nothing to undo or redo.

`DL` starts a dimension instead of a detail line

`DL` is the short form of the linear dimension. To draw detail lines, use **Annotate ▸ Detail ▸ Detail line**, or type `detailline`.

Modelling

«no wall at that point»

A door or a window must be placed in a wall. Click on the wall itself, or give a point on its line at the command line.

A height set at the command line is enormous

At the command line a number without unit is in metres: `set Unconnected Height 2700` makes the wall 2700 m high. Undo with `Ctrl+Z` and write the unit: `2700mm` or `2.7m`. In the Properties palette a number without unit is in the project's units.

Files

Unsaved work is gone after New, Open or Sample project

These commands replace the current project without asking. Save before using them: the asterisk after the project's name in the title bar shows that there are unsaved changes.

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Open… in the browser does not show a file dialog

In the browser, drag the `.bimopen` file into the page: the command line says **Drag a .bimopen file into the window to open it**.

bimopen-cli prints its usage and stops

The arguments are not the ones it expects: it ends with code 2 and shows the usage. A command that fails ends with code 1 and a message saying which line failed. See Command line and MCP.

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